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(54) **SEMICONDUCTOR DEVICE PACKAGE
HAVING THERMAL DISSIPATION FEATURE
AND METHOD THEREFOR**

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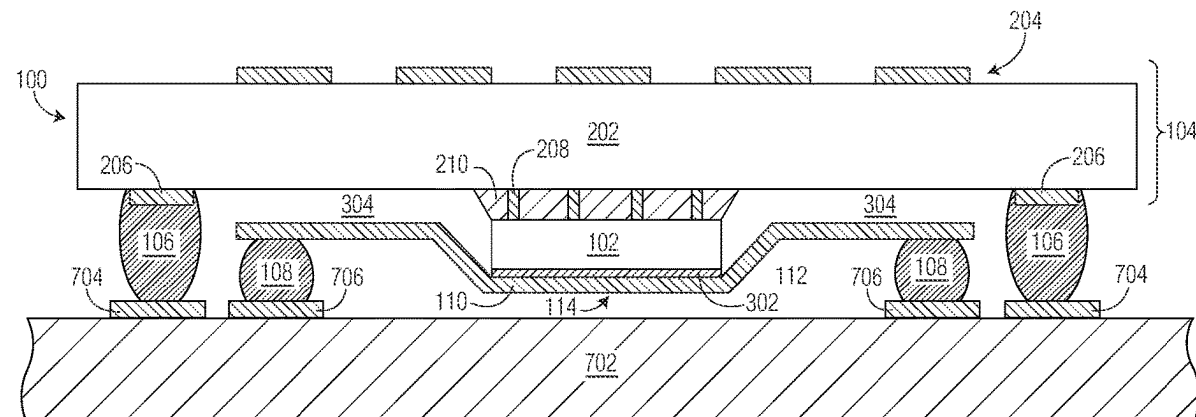
U.S. Appl. No. 17/113,345, filed Dec. 7, 2020, entitled "Semicon-
ductor Device Package Having Thermal Dissipation Feature and
Method Therefor".

Primary Examiner — Duy T Nguyen

(57) **ABSTRACT**

A semiconductor device package having a thermal dissipation feature is provided. The semiconductor device package includes a package substrate. A semiconductor die is mounted on a first surface of the package substrate. A thermal conductive structure including a die pad portion is affixed to the semiconductor die. A limb portion of the thermal conductive structure extends laterally away from the die pad portion and overlaps a portion of the package substrate. A thermal conduction path is formed between the semiconductor die and a distal end of the limb portion.

15 Claims, 12 Drawing Sheets



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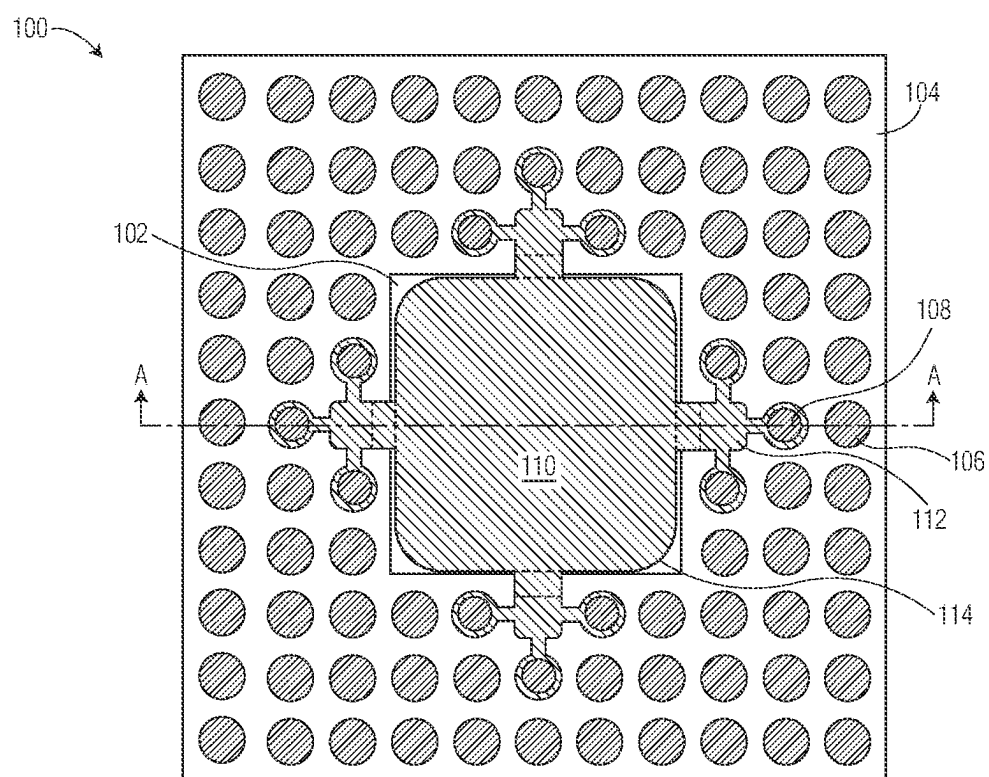


FIG. 1

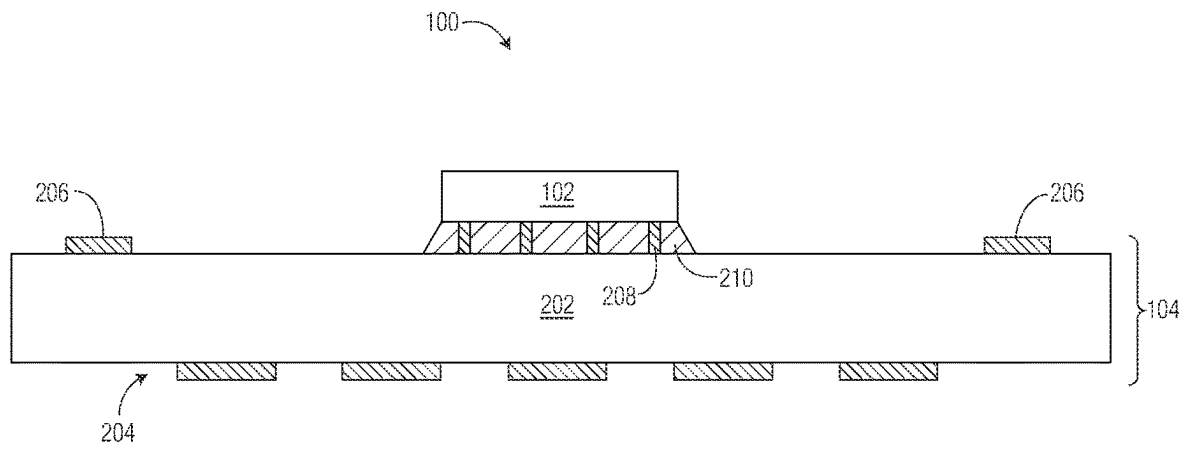


FIG. 2

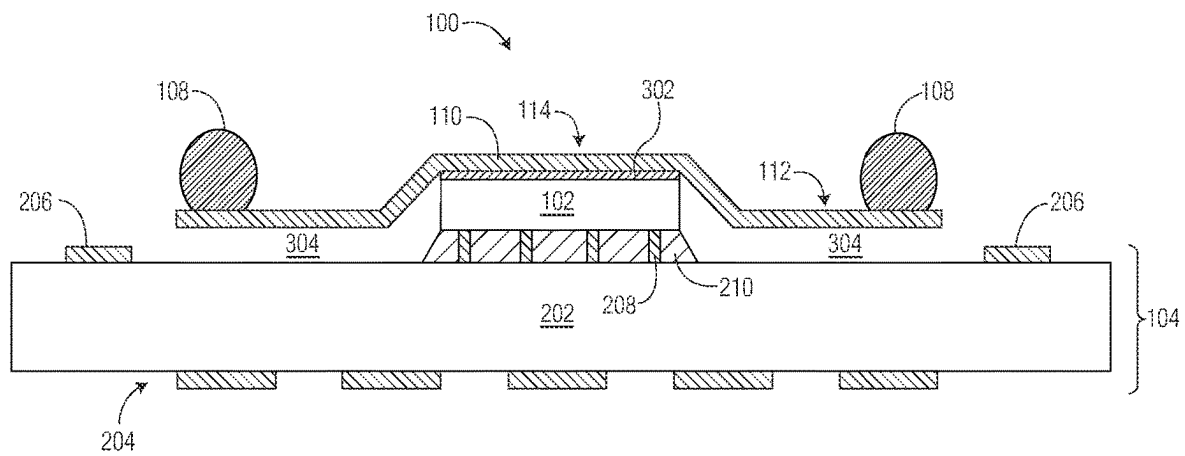


FIG. 3

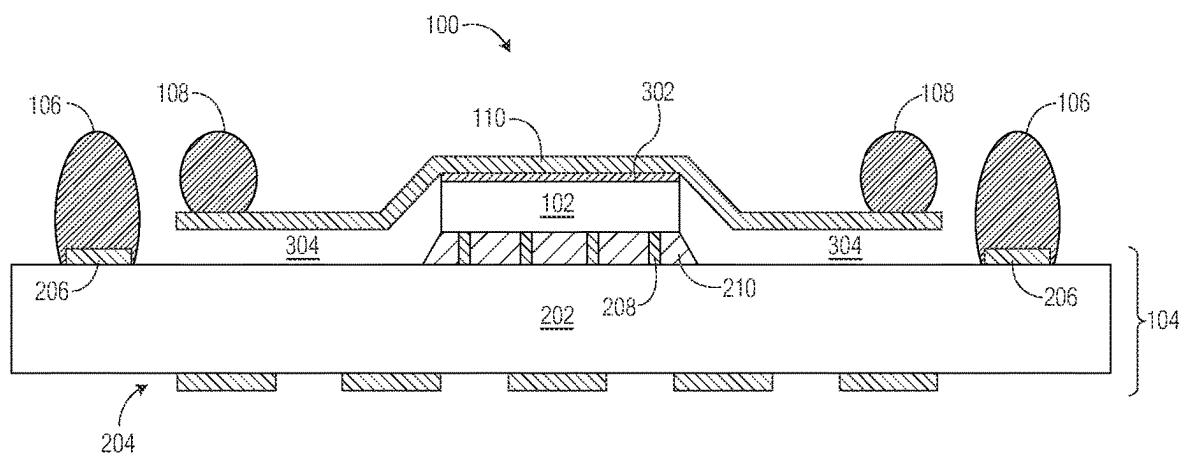


FIG. 4

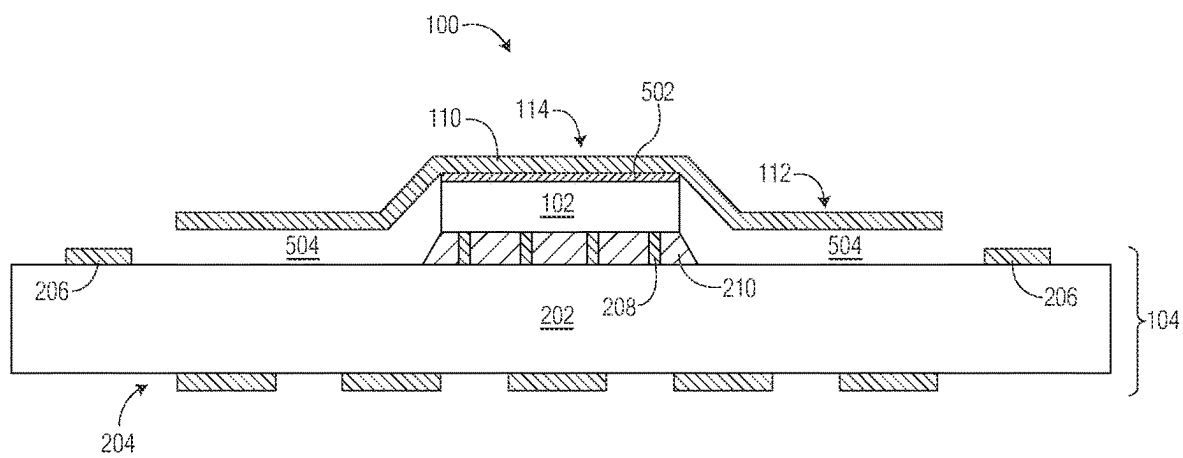


FIG. 5

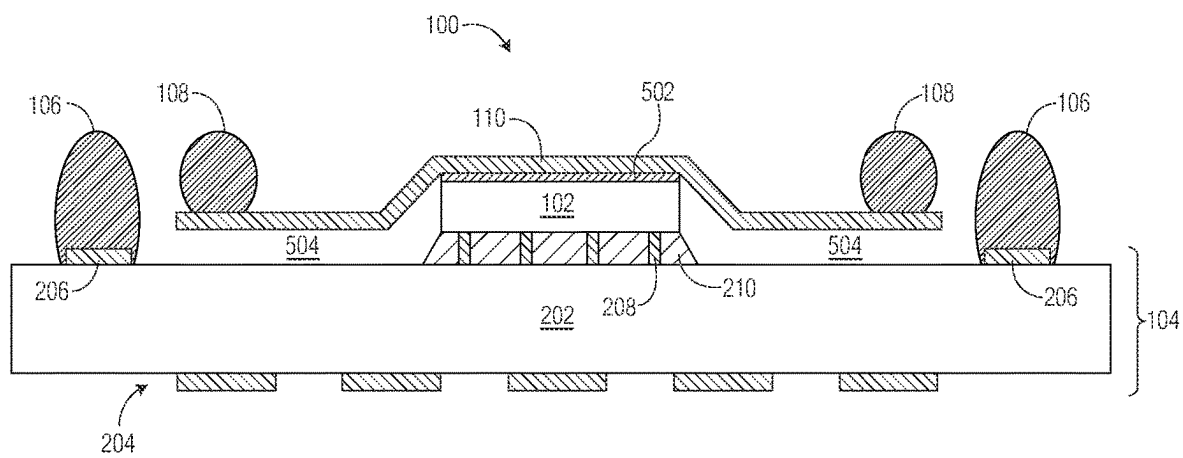


FIG. 6

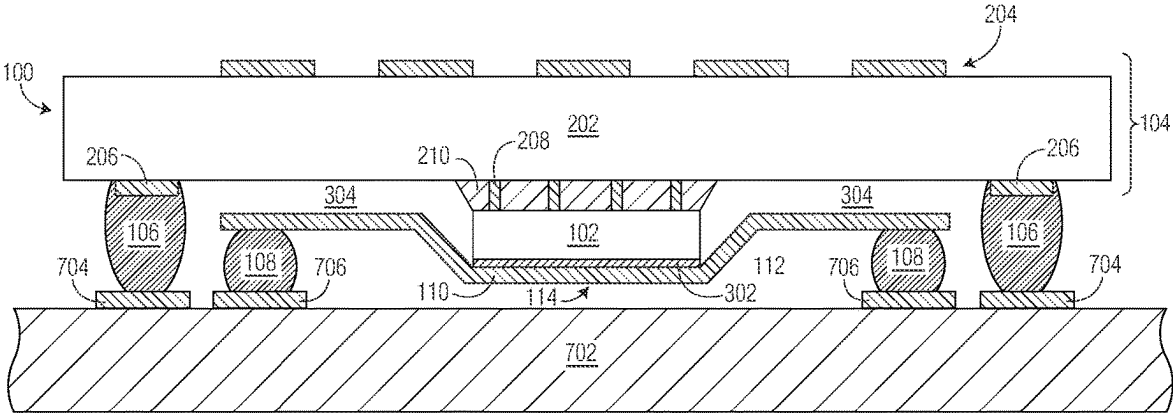


FIG. 7

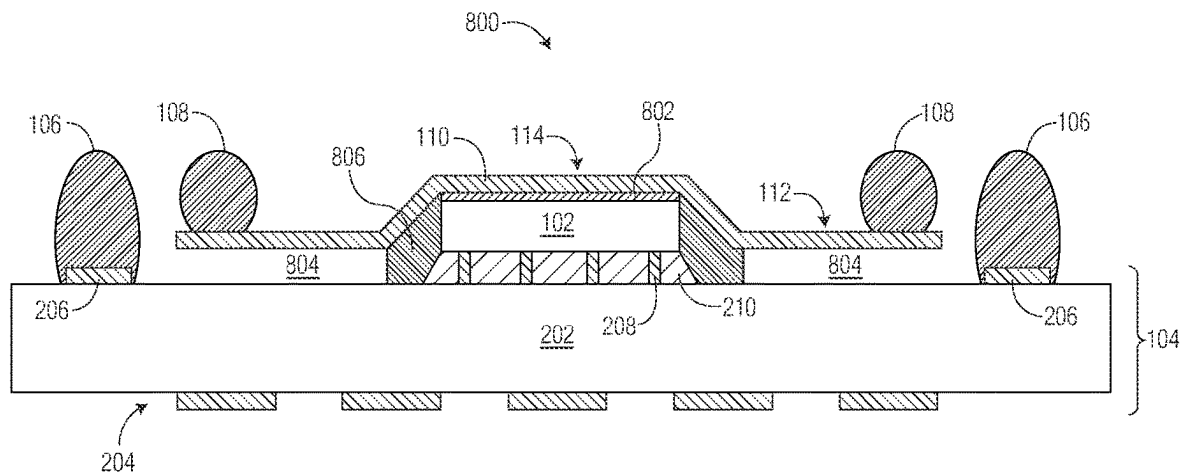


FIG. 8

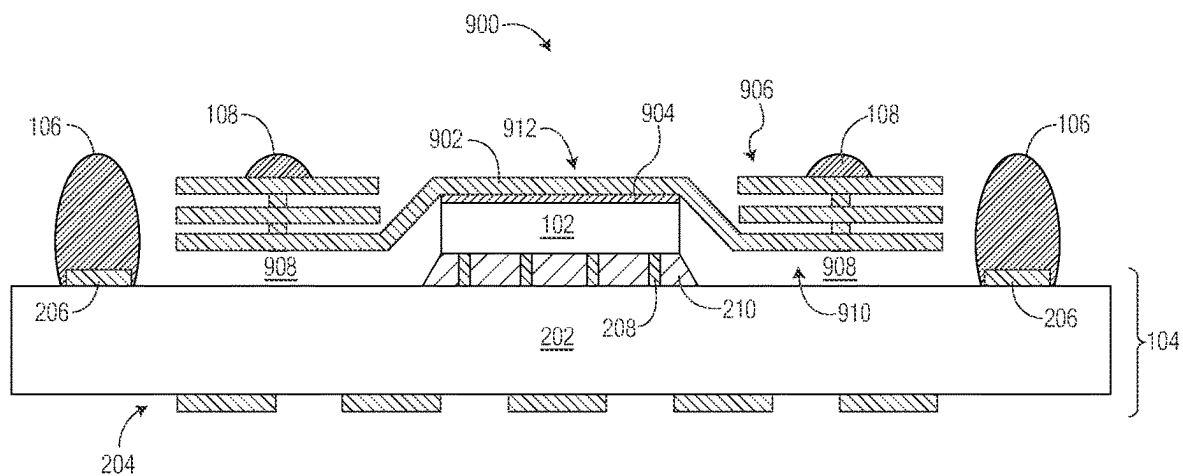


FIG. 9

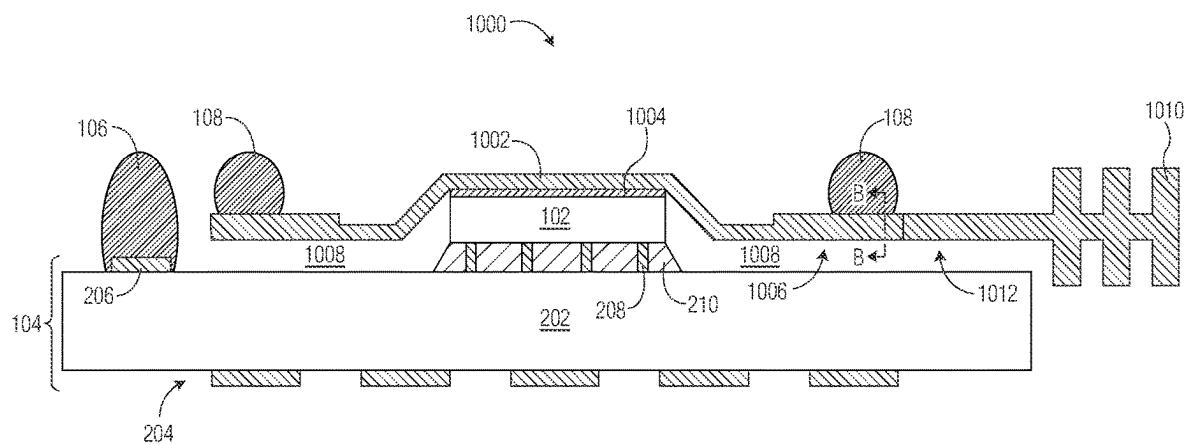
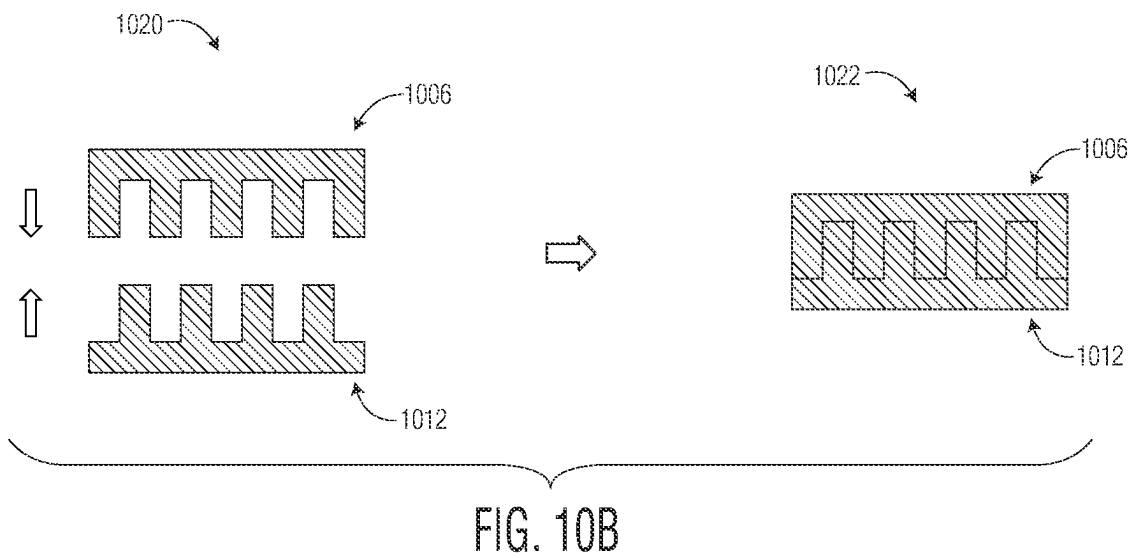


FIG. 10A



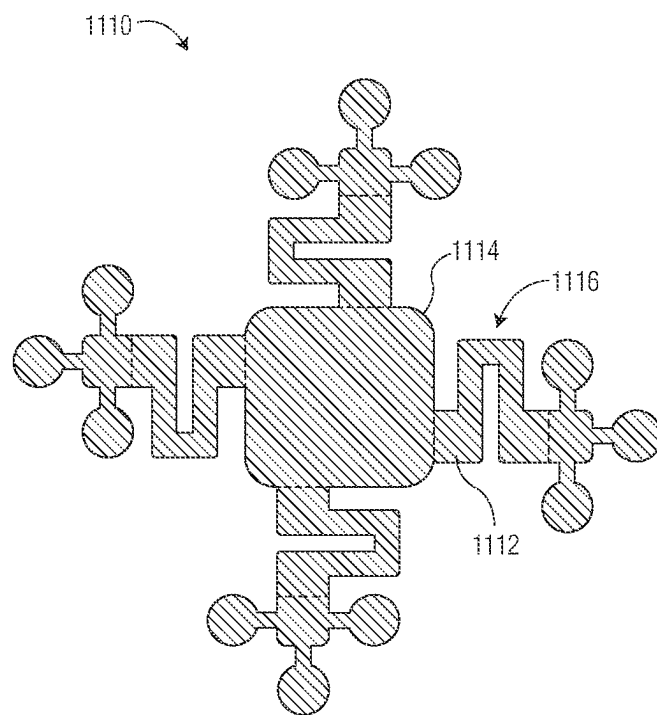


FIG. 11

1

SEMICONDUCTOR DEVICE PACKAGE HAVING THERMAL DISSIPATION FEATURE AND METHOD THEREFOR

BACKGROUND

Field

This disclosure relates generally to semiconductor device packaging, and more specifically, to a semiconductor device package having thermal dissipation feature and method of forming the same.

Related Art

Today, there is an increasing trend to include sophisticated semiconductor devices in products and systems that are used every day. These sophisticated semiconductor devices often have sensitive circuitry and features that can be adversely affected by environmental conditions. For example, the performance of a semiconductor device may be degraded due to excessive heat accumulation. In extreme cases, a semiconductor device may be rendered non-functional due to excessive heat accumulation. Accordingly, there is a need to reduce or eliminate the effects of environmental conditions on such semiconductor devices.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention is illustrated by way of example and is not limited by the accompanying figures, in which like references indicate similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.

FIG. 1 illustrates, in a simplified bottom-side-up plan view, an example semiconductor device package having a thermal dissipation feature in accordance with an embodiment.

FIG. 2 through FIG. 6 illustrate, in simplified cross-sectional views, the example semiconductor device package of FIG. 1 at stages of manufacture in accordance with an embodiment.

FIG. 7 illustrates, in a simplified cross-sectional view, the example semiconductor device package of FIG. 6 as mounted on a printed circuit board (PCB) in accordance with an embodiment.

FIG. 8 illustrates, in a simplified cross-sectional view, an alternative example semiconductor device package having a thermal dissipation feature in accordance with an embodiment.

FIG. 9 illustrates, in a simplified cross-sectional view, another alternative example semiconductor device package having a thermal dissipation feature in accordance with an embodiment.

FIG. 10A and FIG. 10B illustrate, in simplified cross-sectional views, an alternative example semiconductor device package configured with an external heatsink in accordance with an embodiment.

FIG. 11 illustrates, in a simplified bottom-side-up plan view, an example thermal conductive structure with a stress reducing feature in accordance with an embodiment.

DETAILED DESCRIPTION

Generally, there is provided, a semiconductor device package having a thermal dissipation feature. An external thermal conductive structure is affixed to a semiconductor

2

die to form the thermal dissipation structure at a bottom of the package. The external thermal conductive structure includes a die pad portion and a plurality of limb portions which extend over the package substrate. The bottom of the package is attached and interconnected with a printed circuit board by way of conductive connectors. The thermal conductive structure forms a thermal conduction path between the semiconductor die and the printed circuit board. With the thermal conductive structure affixed at the bottom of the package, a top surface of the package remains unobstructed. Accordingly, an antenna structure integrated at the top surface of the package allows for propagation of radio signals, for example, while providing enhanced heat dissipation at the bottom surface of the package. Other features such as a package-on-package interface structure, surface mount technology structure, die bonding structure, shielding structure, and the like may be integrated at the top surface of the package.

FIG. 1 illustrates, in a simplified bottom-side-up plan view, an example semiconductor device package 100 having a thermal dissipation feature in accordance with an embodiment. The package 100 includes a semiconductor die 102 mounted on a package substrate 104 having conductive connectors 106 and 108. The thermal dissipation feature is formed as an external thermal conductive structure 110 affixed on the semiconductor die 102. In this embodiment, the thermal conductive structure 110 includes a die pad portion 114 and four limb portions 112 extending laterally away from the die pad portion and suspended over portions of the package substrate 104. The limb portions 112 include a region formed at a distal end of each limb configured for attachment to a printed circuit board (PCB) by way of conductive connectors 108, for example. Cross-sectional views of examples of the semiconductor device package 100 having a thermal dissipation feature taken along line A-A at stages of manufacture are depicted in FIG. 2 through FIG. 6.

The semiconductor die 102 has an active side (e.g., major side having circuitry, bond pads) and a backside (e.g., major side opposite of the active surface). As depicted in the bottom-side-up plan view of FIG. 1, the active side of the semiconductor die 102 is mounted on the package substrate 104 and interconnected utilizing bond pads (not shown) at the active side, for example. In this embodiment, the semiconductor die 102 is configured in a flip-chip orientation having the active side mounted on the package substrate 104. The semiconductor die 102 may be formed from any suitable semiconductor material, such as silicon, germanium, gallium arsenide, gallium nitride, and the like. Semiconductor die 102 may further include any digital circuits, analog circuits, RF circuits, memory, processor, the like, and combinations thereof at the active surface.

The package substrate 104 has a top surface and a bottom surface (e.g., major surface opposite of the top surface). As depicted in the bottom-side-up plan view of FIG. 1, the active side of the semiconductor die 102 and conductive connectors 106 and 108 are mounted at the bottom surface of the package substrate 104, for example. The conductive connectors 106 may be characterized as input/output and power supply connectors whereas the conductive connectors 108 may be characterized as thermal conduction and heat dissipation path connectors. In some embodiments, the conductive connectors 108 may be configured for connection to a ground power supply, for example, in addition to being part of a thermal conduction and heat dissipation path. The conductive connectors 106 and 108 may be formed in any suitable conductive connector structures such as solder balls, gold studs, copper pillars, and the like. In this embodi-

3

ment, the conductive connectors **106** and **108** are formed as ball connectors, for example, and arranged in a ball grid array (BGA). The package substrate **104** is formed as a multi-layer laminate structure having conductive features (e.g., metal traces, pads) separated by non-conductive material (e.g., FR-4). In this embodiment, the package substrate **104** is characterized as a chip scale package (CSP) type package substrate.

The thermal conductive structure **110** affixed to the semiconductor die **102** forms a thermal conduction and heat dissipation path between the semiconductor die **102** and conductive connectors **108**. The thermal conductive structure **110** may be formed from suitable thermal conductive materials such as copper, silver, nickel, aluminum, and the like. The thermal conductive structure **110** including the die pad portion **114** and the limb portions **112** may be formed from a stamped or etched metal sheet. For example, the thermal conductive structure **110** may be manufactured in a manner consistent with the manufacture of a package lead-frame. The thermal conductive structure **110** may be affixed to the semiconductor die **102** by way of a thermally conductive adhesive or solder, for example. The thermal conductive structure **110** may be affixed to the semiconductor die **102** of packages while in an array (e.g., strip) form or in singulated form.

FIG. 2 through FIG. 6 illustrate, in simplified cross-sectional views, the example semiconductor device package **100** taken along line A-A of FIG. 1 at stages of manufacture in accordance with an embodiment.

FIG. 2 illustrates, in a simplified bottom-side-up cross-sectional view, the example semiconductor device package **100** at a stage of manufacture in accordance with an embodiment. At this stage, the package **100** includes the active side of the semiconductor die **102** interconnected with the bottom surface of the package substrate **104** by way of conductive die connectors **208** and underfilled with an epoxy material **210**. The die connectors **208** may be any suitable die connector structure such as solder bumps, gold studs, copper pillars, and the like. Features such as bond pads on the semiconductor die **102** and corresponding pads on the package substrate **104** are not shown for illustration purposes.

The package substrate **104** is formed as a multi-layer laminate structure having conductive features separated by a non-conductive material. In this embodiment, the package substrate **104** includes connector pads **206** located at the bottom surface and a conductive structure **204** integrated at the top surface of the package substrate **104**. In this embodiment, the integrated conductive structure **204** is characterized as an antenna structure configured for propagation of radio frequency (RF) signals such as a radar, Wi-Fi, UWB, 5G and 6G signals, for example. In other embodiments, the integrated conductive structure **204** may include features such as a package-on-package (PoP) interface structure, surface mount technology structure, die bonding structure, shielding structure, and the like integrated at the top surface of the package substrate **104**. The connector pads **206** and the integrated conductive structure **204** are separated by a non-conductive material **202**. The connector pads **206** and the integrated conductive structure **204** may be formed from a suitable metal such as copper, for example. Interconnect features such as conductive vias and traces embedded in the package substrate **104** are not shown for illustration purposes.

FIG. 3 illustrates, in a simplified bottom-side-up cross-sectional view, the example semiconductor device package **100** at a subsequent stage of manufacture in accordance with

4

an embodiment. At this stage, the package **100** includes the thermal conductive structure **110** is mounted on the semiconductor die **102** and conductive connectors **108** are affixed on thermal conductive structure **110**. In this embodiment, the thermal conductive structure **110** includes the die pad portion **114** affixed to the backside of the semiconductor die **102** by way of a thermal conductive bonding material **302** (e.g., thermally conductive adhesive, solder, paste, epoxy, die attach film). The limb portions **112** extend laterally away from the die pad portion **114** and are downset forming a gap **304** between the limb portions **112** and the bottom surface of the package substrate **104**. The gap **304** may be an air gap or filled with a low modulus material (e.g., for mechanical stability) or a thermally conductive material (e.g., for additional thermal conduction, dissipation), for example. The conductive connectors **108** are affixed at the distal ends of the limb portions **112**. A region at the distal ends of the limb portions **112** may include features such as plating and/or dimples conducive for attachment of conductive connectors, for example. The conductive connectors **108** be any suitable conductive connector structures such as solder balls, gold studs, copper pillars, and the like. In some embodiments, conductive connectors **108** may be formed with enhanced thermal conductivity materials and/or structures.

FIG. 4 illustrates, in a simplified bottom-side-up cross-sectional view, the example semiconductor device package **100** at a subsequent stage of manufacture in accordance with an embodiment. At this stage, the package **100** includes the thermal conductive structure **110** mounted on the semiconductor die **102** and conductive connectors **106** affixed on connector pads **206**. After the thermal conductive structure **110** is mounted on the semiconductor die **102**, the conductive connectors **106** are affixed on the respective connector pads **206**. In this embodiment, the conductive connectors **106** are coupled to provide input/output and power supply connections to the semiconductor die **102** by way of the connector pads **206**.

FIG. 5 and FIG. 6 illustrate, in simplified bottom-side-up cross-sectional views, the example semiconductor device package **100** taken along line A-A of FIG. 1 at alternative stages of manufacture in accordance with an embodiment. The embodiments depicted in FIG. 5 and FIG. 6 are subsequent to the stage of manufacture depicted in FIG. 2 and are alternatives to the embodiments depicted in FIG. 3 and FIG. 4.

FIG. 5 illustrates, in a simplified bottom-side-up cross-sectional view, the example semiconductor device package **100** at an alternative stage of manufacture in accordance with an embodiment. At this stage, the package **100** includes the thermal conductive structure **110** mounted on the semiconductor die **102**. In this embodiment, the thermal conductive structure **110** includes the die pad portion **114** affixed to the backside of the semiconductor die **102** by way of a thermal conductive bonding material **502**. The limb portions **112** extend laterally away from the die pad portion **114** and are downset forming a gap **504** between the limb portions **112** and the bottom surface of the package substrate **104**. A region at the distal ends of the limb portions **112** are configured for attachment to a PCB by way of conductive connectors, for example.

FIG. 6 illustrates, in a simplified bottom-side-up cross-sectional view, the example semiconductor device package **100** at an alternative stage of manufacture in accordance with an embodiment. At this stage, the package **100** includes the thermal conductive structure **110** mounted on the semiconductor die **102** and conductive connectors **106** and **108** affixed on respective connector pads **206** and the thermal

5

conductive structure **110**. After the thermal conductive structure **110** is mounted on the semiconductor die **102**, the conductive connectors **106** are affixed on the connector pads **206** and the conductive connectors **108** are affixed at the distal ends of the limb portions **112**. In this embodiment, the conductive connectors **106** are coupled to provide input/output and power supply connections to the semiconductor die **102** by way of the connector pads **206**.

FIG. 7 illustrates, in a simplified cross-sectional view, the example semiconductor device package **100** of FIG. 4 (and FIG. 6) as mounted on a PCB **702** in accordance with an embodiment. In this embodiment, the package **100** is oriented (e.g., flipped), attached and interconnected to the PCB **702** by way of the conductive connectors **106** and **108**. The conductive connectors **106** and **108** are affixed to respective pads **704** and **706** of the PCB **702**. In this embodiment, a thermal conduction path from the semiconductor die **102** is extended to the PCB by way of the conductive connectors **108** and the respective pads **706**. In some embodiments, an additional thermal conduction path may be formed between the semiconductor die **102** and a pad (not shown) formed on the PCB directly below the die pad portion **114**. Because the thermal conduction path is formed at the bottom side of the package **100**, the integrated conductive structure **204** at the top surface of the package substrate remains unobstructed.

FIG. 8 illustrates, in a simplified bottom-side-up cross-sectional view, an alternative example semiconductor device package **800** having a thermal dissipation feature in accordance with an embodiment. At this stage, the package **800** includes a thermal conductive fillet **806** formed at the semiconductor die **102** to package substrate **104** transition. A thermally conductive material (e.g., thermal epoxy, paste) is dispensed around a perimeter of the semiconductor die **102** such that a fillet is formed at the semiconductor die **102** to package substrate **104** transition. The thermally conductive material may be dispensed before or after the thermal conductive structure **110** is mounted on the semiconductor die **102** by way of a thermal conductive bonding material **802**. The thermal conductive fillet **806** is configured and arranged to directly contact sidewalls of the semiconductor die **102** and the thermal conductive structure **110**. In this arrangement, the limb portions **112** are further supported, and additional heat dissipation paths are formed from the sidewalls of the semiconductor die **102**. A gap **804** remains between the limb portions **112** and the package substrate **104**. In some embodiments, the gap **804** may be an air gap or filled with a low modulus material (e.g., for additional mechanical stability) or a thermally conductive material (e.g., for additional thermal conduction, dissipation), for example. Because a thermal conduction path is formed (e.g., from the semiconductor die **102** to the conductive connector **108**) at the bottom side of the package **800**, the integrated conductive structure **204** at the top surface of the package substrate remains unobstructed.

FIG. 9 illustrates, in a simplified bottom-side-up cross-sectional view, another alternative example semiconductor device package **900** having a thermal dissipation feature in accordance with an embodiment. At this stage, the package **900** includes a thermal conductive structure **902** mounted on the semiconductor die **102** and conductive connectors **106** and **108** affixed on respective connector pads **206** and the thermal conductive structure **902**. In this embodiment, the thermal conductive structure **902** includes a die pad portion **912** affixed to the backside of the semiconductor die **102** by way of a thermal conductive bonding material **904**. Limb portions **910** of the thermal conductive structure **902** extend laterally away from the die pad portion **912** and are downset

6

forming a gap **908** between the limb portions **910** and the package substrate **104**. In this embodiment, the thermal conductive structure **902** includes example thermal radiation fins **906** formed on the limb portions **910** and configured for additional heat dissipation. The thermal radiation fins **906** may be formed from similar materials (e.g., copper) as the thermal conductive structure **902**, and formed in various suitable shapes and orientations. In some embodiments, the outer-most surface of the thermal radiation fins **906** may be configured for direct contact with a PCB using a thermal paste, for example. In some embodiments, the outer-most surface of the thermal radiation fins **906** may be configured for a direct solder connection with a PCB without using conductive connectors, for example. Because a thermal conduction path is formed (e.g., from the semiconductor die **102** to the conductive connector **108**) at the bottom side of the package **800**, the integrated conductive structure **204** at the top surface of the package substrate remains unobstructed.

FIG. 10A illustrates, in a simplified bottom-side-up cross-sectional view, an alternative example semiconductor device package **1000** configured with an external heatsink structure **1010** in accordance with an embodiment. At this stage, the package **1000** includes a thermal conductive structure **1002** mounted on the semiconductor die **102** and attached to the external heatsink structure **1010**. Conductive connectors **106** and **108** are affixed on respective connector pads **206** and the thermal conductive structure **1002**. Cross-sectional views of an example interlocking connection structure taken along line B-B is depicted in FIG. 10B.

In the embodiment depicted in FIG. 10A, the thermal conductive structure **1002** includes a die pad portion affixed to the backside of the semiconductor die **102** by way of a thermal conductive bonding material **1004**. Limb portions of the thermal conductive structure **1002** extend laterally away from the die pad portion and are downset forming a gap **1008** between the limb portions and the package substrate **104**. In this embodiment, the thermal conductive structure **1002** includes an example interlocking connection structure portion **1006** formed at the distal ends of the limb portions and configured for attachment of the external heatsink structure **1010** for additional heat dissipation. When the external heatsink structure **1010** is joined with the thermal conductive structure **1002**, a continuous thermal conduction path from the semiconductor die **102** to the external heatsink structure **1010** is formed. The external heatsink structure **1010** may be formed from a suitable metal (e.g., copper) having any suitable structure configured for heat dissipation. In this embodiment, a thermal conduction path also extends to the PCB by way of the conductive connectors **108** affixed on the thermal conductive structure **1002**. Because the thermal conduction paths are formed at the bottom side of the package **1000**, the integrated conductive structure **204** at the top surface of the package substrate **104** remains unobstructed.

The example interlocking connection structure taken along line B-B is shown in more detail in FIG. 10B. The connection structure portion **1006** of the thermal conductive structure **1002** and a corresponding interlocking connection structure portion **1012** of the external heatsink **1010** are depicted in an unmated arrangement **1020**. The top side of the connection structure portion **1006** is configured for attachment to a PCB by way of conductive connectors **108**, for example. The bottom side interlocking features may serve as a heatsink, for example, when an external heatsink is not attached. The corresponding interlocking connection structure portion **1012** formed on the external heatsink

structure **1010** is configured for mating with the connection structure portion **1006**. The connection structure portion **1006** interlocked with the corresponding connection structure portion **1012** is depicted in a mated arrangement **1022**. The example interlocking connection structure, when in the mated arrangement **1022**, ensures a sufficient thermal conduction. A thermal grease or paste applied on interlocking features may further increase the thermal conductivity of the interlocking connection structure.

FIG. **11** illustrates, in a simplified bottom-side-up plan view, an example thermal conductive structure **1110** with a stress reducing feature in accordance with an embodiment. The thermal conductive structure **1110** depicted in FIG. **11** is an alternative to the thermal conductive structure **110** depicted in FIG. **1**. The thermal conductive structure **1110** is configured for attachment on a semiconductor die for improved heat dissipation. In this embodiment, the thermal conductive structure **1110** includes a die pad portion **1114** and four limb portions **1112** extending laterally away from the die pad portion. The limb portions **1112** include a region formed at a distal end of each limb configured for attachment to a PCB by way of conductive connectors, for example. The thermal conductive structure **1110**, when affixed to the semiconductor die, forms a thermal conduction and heat dissipation path between the semiconductor die and the conductive connectors.

In this embodiment, each limb portion **1112** includes a U-shaped stress relief feature **1116** formed between the die pad portion **1114** and the distal end of the limb portion **1112**. The stress relief feature **1116** is configured to allow more flexibility in the limb portions **1112** after being affixed to a PCB, for example. The stress relief feature **1116** also provides more surface area for heat dissipation. The thermal conductive structure **1110** may be formed from suitable thermal conductive materials such as copper, silver, nickel, aluminum, and the like. The thermal conductive structure **1110** including the die pad portion **1114** and the limb portions **1112** may be formed from a stamped or etched metal sheet. For example, the thermal conductive structure **1110** may be manufactured in a manner consistent with the manufacture of a package leadframe.

Generally, there is provided, a semiconductor device package including a package substrate having a first major surface and a second major surface; a semiconductor die mounted on the first major surface of the package substrate; and a thermal conductive structure including a die pad portion affixed to the semiconductor die and a limb portion extending laterally away from the die pad portion, the limb portion located substantially over a portion of the package substrate, the thermal conductive structure configured to form a thermal conduction path between the semiconductor die and a distal end of the limb portion. A region at the distal end of the limb portion may be configured for attachment to a printed circuit board by way of a first conductive connector. The semiconductor device package may further include a connector pad at the first surface of the package substrate, the connector pad configured for attachment to the printed circuit board by way of a second conductive connector. The thermal conductive structure including the die pad portion and the limb portion may be formed from a stamped or etched metal sheet. The semiconductor device package may further include a thermal conductive fillet formed at the semiconductor die to package substrate transition, the fillet directly contacting sidewalls of the semiconductor die and the thermal conductive structure. The semiconductor device package may further include an antenna structure integrated at the second surface of the package substrate. The thermal

conductive structure may further include one or more thermal conductive fins attached to the limb portion of the thermal conductive structure. The limb portion of the thermal conductive structure may include a stress relief feature configured to reduce stress when the limb portion is attached to a printed circuit board. The distal end of the limb portion of the thermal conductive structure may be further configured for attachment of an external heatsink structure.

In another embodiment, there is provided, a method including mounting a semiconductor die on a first surface of a package substrate; providing a thermal conductive structure including a die pad portion and a limb portion extending laterally away from the die pad portion; and affixing the die pad portion of the thermal conductive structure to the semiconductor die, the limb portion suspended over a portion of the package substrate. The method may further include affixing a first conductive connector at a distal end of the limb portion of the thermal conductive structure, the first conductive connector configured for attachment to a printed circuit board. The method may further include affixing a second conductive connector on a connector pad located at first surface of the package substrate, the second conductive connector configured for attachment to the printed circuit board. The package substrate may include an antenna structure integrated at a second surface of the package substrate. The method may further include forming a thermal conductive fillet at the semiconductor die to package substrate transition, the fillet directly contacting sidewalls of the semiconductor die and the thermal conductive structure. The limb portion of thermal conductive structure may be downset relative to the die pad portion of the thermal conductive structure. The affixing the die pad portion of the thermal conductive structure to the semiconductor die may include affixing the die pad portion to the semiconductor die by way of a thermally conductive adhesive or solder.

In yet another embodiment, there is provided, a semiconductor device package including a package substrate having a first major surface and a second major surface; a semiconductor die having an active side and a backside, the active side mounted on the first major surface of the package substrate; and a thermal conductive structure having a die pad portion affixed to the backside of the semiconductor die and a limb portion extending laterally away from the die pad portion, the limb portion located substantially over a portion of the package substrate, a gap formed between the limb portion and the first major surface of the package substrate. The semiconductor device package may further include a first conductive connector affixed at a distal end of the limb portion and configured for attachment to a printed circuit board, a thermal conduction path formed between the semiconductor die and the first conductive connector. The semiconductor device package may further include a second conductive connector affixed on a connector pad at first surface of the package substrate, the second conductive connector configured for attachment to the printed circuit board. The first conductive connector and the second conductive connector may be characterized as balls of a ball grid array (BGA) package.

By now, it should be appreciated that there has been provided a semiconductor device package having a thermal dissipation feature. An external thermal conductive structure is affixed to a semiconductor die to form the thermal dissipation structure at a bottom of the package. The external thermal conductive structure includes a die pad portion and a plurality of limb portion which extend over the package substrate. The bottom of the package is attached and inter-

connected with a printed circuit board by way of conductive connectors. The thermal conductive structure forms a thermal conduction path between the semiconductor die and the printed circuit board. With the thermal conductive structure affixed at the bottom of the package, a top surface of the package remains unobstructed. Accordingly, an antenna structure integrated at the top surface of the package allows for propagation of radio signals, for example, while providing enhanced heat dissipation at the bottom surface of the package. Other features such as a package-on-package interface structure, surface mount technology structure, die bonding structure, shielding structure, and the like may be integrated at the top surface of the package.

The terms “front,” “back,” “top,” “bottom,” “over,” “under” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the invention described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

Although the invention is described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present invention. Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

Furthermore, the terms “a” or “an,” as used herein, are defined as one or more than one. Also, the use of introductory phrases such as “at least one” and “one or more” in the claims should not be construed to imply that the introduction of another claim element by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim element to inventions containing only one such element, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an.” The same holds true for the use of definite articles.

Unless stated otherwise, terms such as “first” and “second” are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements.

What is claimed is:

1. A semiconductor device package comprising:

a package substrate having a first major surface and a second major surface;

a semiconductor die mounted on the first major surface of the package substrate; and

a thermal conductive structure including a die pad portion affixed to the semiconductor die and a limb portion extending laterally away from a first edge of the die pad portion in a first direction parallel to the first major surface, the limb portion having a width less than a width of the die pad portion at the first edge measured in a second direction that is parallel to the first major surface and perpendicular to the first direction, the limb portion overhanging a portion of the package substrate; wherein the thermal conductive structure is configured to form a thermal conduction path between the semicon-

ductor die and a distal end of the limb portion that is suspended directly above the first major surface of the package substrate;

wherein a region at the distal end of the limb portion is configured for attachment to a printed circuit board by way of a first conductive connector disposed on a side of the limb portion that faces away from the package substrate; and

wherein the semiconductor device package further comprises a connector pad at the first major surface of the package substrate, the connector pad configured for attachment to the printed circuit board by way of a second conductive connector.

2. The semiconductor device package of claim 1, wherein the thermal conductive structure including the die pad portion and the limb portion is formed from a stamped or etched metal sheet.

3. The semiconductor device package of claim 1, further comprising a thermal conductive fillet formed at the semiconductor die to package substrate transition, the fillet directly contacting sidewalls of the semiconductor die and the thermal conductive structure.

4. The semiconductor device package of claim 1, further comprising an antenna structure integrated at the second surface of the package substrate.

5. The semiconductor device package of claim 1, wherein the thermal conductive structure further comprises one or more thermal conductive fins attached to the limb portion of the thermal conductive structure.

6. The semiconductor device package of claim 1, wherein a width of the thermal conductive structure between the first edge of the die pad portion and the distal end of the limb portion is equal to the width of the limb portion.

7. A semiconductor device package, comprising:

a package substrate having a first major surface and a second major surface;

a semiconductor die mounted on the first major surface of the package substrate;

a thermal conductive structure including a die pad portion affixed to the semiconductor die and a limb portion extending laterally away from a first edge of the die pad portion in a first direction parallel to the first major surface, the limb portion having a width less than a width of the die pad portion measured in a second direction that is parallel to the first major surface and perpendicular to the first direction, the limb portion overhanging a portion of the package substrate;

a first electrical connector pad on the first major surface configured to allow the first major surface to be bonded to a printed circuit board or other carrier;

wherein the thermal conductive structure is configured to form a thermal conduction path between the semiconductor die and a distal end of the limb portion that is suspended directly above the package substrate; and

wherein the limb portion of the thermal conductive structure includes a stress relief feature configured to reduce stress when the limb portion is attached to the printed circuit board or other carrier.

8. A method comprising:

mounting a semiconductor die on a first surface of a package substrate;

providing a thermal conductive structure including a die pad portion and a limb portion, the limb portion extending laterally away from a first edge of the die pad portion in a first direction parallel to the first surface, the limb portion having a width less than a width of the die pad portion at the first edge measured in a second

11

- direction that is parallel to the first major surface and perpendicular to the first direction; and
- affixing the die pad portion of the thermal conductive structure to the semiconductor die, the limb portion suspended directly over a portion of the package substrate at a distal end of the limb portion;
- wherein the method further comprises affixing a first conductive connector at a distal end of the limb portion of the thermal conductive structure, the first conductive connector configured for attachment to a circuit board; and
- affixing a second conductive connector on a connector pad located at first surface of the package substrate, the second conductive connector configured for attachment to the circuit board.
9. The method of claim 8, wherein the package substrate includes an antenna structure integrated at a second surface of the package substrate.
10. The method of claim 8, further comprising forming a thermal conductive fillet at the semiconductor die to package substrate transition, the fillet directly contacting sidewalls of the semiconductor die and the thermal conductive structure.
11. The method of claim 8, wherein the limb portion of thermal conductive structure is downset relative to the die pad portion of the thermal conductive structure.
12. The method of claim 8, wherein affixing the die pad portion of the thermal conductive structure to the semiconductor die includes affixing the die pad portion to the semiconductor die by way of a thermally conductive adhesive or solder.
13. A semiconductor device package comprising:
a package substrate having a first major surface and a second major surface;

12

- a semiconductor die having an active side and a backside, the active side mounted on the first major surface of the package substrate;
- a thermal conductive structure having a die pad portion affixed to the backside of the semiconductor die and a limb portion extending laterally away from a first edge of the die pad portion in a first direction that is parallel to the first major surface, the limb portion having a width less than a width of the die pad portion at the first edge measured in a second direction that is parallel to the first major surface and perpendicular to the first direction, the limb portion suspended directly over a portion of the package substrate, a gap formed between the limb portion and the first major surface of the package substrate;
- a first conductive connector affixed at a distal end of the limb portion on a surface of the limb portion that faces away from the first major surface and configured for attachment to a circuit board; and
- a second conductive connector affixed on a connector pad at the first major surface of the package substrate, the second conductive connector configured for attachment to the circuit board;
- wherein the entire limb portion of the thermal conductive structure is separated from the package substrate by the gap.
14. The semiconductor device package of claim 13, wherein a thermal conduction path is formed between the semiconductor die and the first conductive connector.
15. The semiconductor device package of claim 13, wherein the first conductive connector and the second conductive connector are characterized as balls of a ball grid array (BGA) package.

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